

DEGREE PROJECT IN TECHNOLOGY,
SECOND CYCLE, 30 CREDITS
STOCKHOLM, SWEDEN 2026



Tube neural orbit predictive control in the vicinity of asteroids with uncertain dynamics

Master's thesis report

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Acknowledgments

I would like to express my sincere gratitude to my examiner, Pr. Christer Fuglesang, and to my supervisors, Dr. Erwin Mooij, Dr. Bart Root, Pedro Roque, and Alessio Russo, for their guidance. Without them, any of this work would have been possible. Erwin and Bart gave me the incredible opportunity to be part of their research group at TU Delft for a period of 6 months and introduced me to the fascinating topic of Guidance, Navigation, and Control systems around asteroids. They made a lot of valuable suggestions during the weekly meetings and gave me enough freedom to take directions on my own. Pedro and Alessio were always there to assist with the supervision of my work. Pedro advised me relevant papers to read especially on neural and robust model predictive controllers. Alessio was always willing to help me with the learning part of my thesis and the implementation of my work.

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Abstract

This thesis presents a Neural Tube Model Predictive Controller (MPC) for orbit control around asteroids. The complex and partially known gravity field of asteroids is modeled by a Physics-Informed Neural Network (PINN) trained on a high-resolution polyhedron model. This PINN gravity model, which offers improvements in computational efficiency over past analytic representations, is integrated as part of the dynamics within a nominal MPC pipeline. An online update of the PINN gravity model is investigated. A Tube MPC formulation with a Boundary Layer Sliding Mode Ancillary Controller is proposed to ensure robustness against unmodelled gravity and solar disturbances. The performances of the designed controller are illustrated in a realistic mission scenario on an unstable polar orbit around 433 Eros.

Sammanfattning

Denna avhandling presenterar en Neural Tube Model Predictive Controller (MPC) för kontroll av omloppsbana runt asteroider. Det komplexa och delvis kända gravitationsfältet för asteroider är modellerat av ett Physics-Informed Neural Network (PINN) tränat på en högupplöst polyedermodell. Denna PINN gravitationsmodell, som erbjuder förbättringar i beräkningseffektivitet jämfört med tidigare analytiska representationer, är integrerad som en del av dynamiken i en nominell MPC-pipeline. En onlineuppdatering av PINN gravitationsmodellen undersöks. En Tube-MPC-formulering med en Boundary Layer Sliding Mode Ancillary Controller föreslås för att säkerställa robusthet mot omodellerad gravitation och solstörningar. Den designade styrenhetens prestanda illustreras i ett realistiskt scenario på en instabil polär bana runt 433 Eros.

Acronyms

MPC Model Predictive Control

NLP Nonlinear Programming Problem

OCP Optimal Control Problem

PINN Physics-Informed Neural Network

QP Quadradic Programming

SQP Sequential Quadradic Programming

1 Introduction

Future asteroid exploration can enable a deeper understanding of the early solar system and the origin of life [15]. Currently, the Lucy mission to five Jupiter Trojan asteroids at L4 [20] and the Psyche mission to the rare metallic asteroid 16 Psyche [19] are underway.

Mission design to non-visited asteroids is challenging because the spacecraft needs to travel safely in a partially unknown and complex dynamic environment with a limited quantity of fuel. The gravity field of asteroids is often highly in-homogeneous due to their irregular shape and heterogeneous mass distribution, and hard to predict. In addition, the solar radiation pressure, which is difficult to model, can be a significant disturbance given the weak gravitational attraction of asteroids due to their small size. These perturbations yield large and often unpredictable deviations from the Keplerian dynamics of spherical bodies that can lead the spacecraft to escape trajectories or collide with the asteroid [28].

To prevent this, trajectory-correcting maneuvers operated from Earth are performed to guarantee the tracking of pre-computed reference orbits. However the communication delays of several minutes between the spacecraft and Earth constraints drastically the permissible limits for uncertainty and consequently restrict the range of possible orbital operations. To eliminate the ground segment dependency, recent research started to investigate autonomous and robust tracking control systems.

Several tracking methods have been proposed in the literature for close proximity orbital operations. Many of them are based on tracking error minimization without optimizing the control efforts [33] [35] [29].

A control method that can simultaneously minimize tracking error and control effort is Model Predictive Control (MPC)[23]. MPC is a control technique that recursively predicts the future behavior of the control system over a finite time horizon and computes an optimal control input that, while ensuring the satisfaction of constraints, minimizes a user-chosen cost function.

Recent years have seen an increasing interest in applying MPC for autonomous spacecraft missions. For example, it has been employed for low-earth station keeping [34] and spacecraft Rendez-Vous [4].

MPC is a model-based controller that requires an accurate and computationally efficient model of the system to be integrated into an onboard system. However, the gravity acceleration, which is the main force acting on a spacecraft placed in a low orbit, is usually largely unknown and very complex to model.

The spherical harmonic model is the most popular approach to model the gravity field of celestial bodies for orbital operations [3]. It is a differentiable model, whose accuracy and complexity scales with the number of considered harmonics. To have a model compatible with a nominal MPC pipeline, Sanchez et al. [26] proposed to model the gravity field of asteroids by a truncated fourth-order spherical harmonic model with unknown coefficients which are estimated online through an unscented Kalman filter (UKF). This approach, however, suffers from drawbacks: (1) the gravity field of such bodies is characterized by high-order gravity field perturbations that cannot be accurately represented by a computationally efficient low-rank harmonics model. For example, to accurately represent perturbing features such as craters, one may require hundreds of thousands of harmonics.

[12]; (2) the high-order harmonic coefficients can be difficult to regress online, requiring strict sampling requirements for observability [9].

Martin and Schaub [11] proposed recently an alternative to the spherical harmonic model by introducing the Physics-Informed Neural Network (PINN) gravity model. PINNs are able to learn pre-existing gravity models using the modeling power of neural networks, be differentiated through Automatic Differentiation, and be evaluated online at a low computational cost. This machine-learning approach allows heavy models, such as polyhedral gravity models, to be integrated into an onboard framework. Moreover, preliminary research [13] has shown that a PINN can be trained online, improving the accuracy of the gravity model with fewer and noisier measurements than the spherical harmonic model.

Independently of the chosen gravity model, the full dynamic model of the spacecraft will always suffer from inaccuracies compared to the real gravity and solar perturbations that may affect the spacecraft. These inaccuracies should be taken into account while designing the final controller. A popular extension of MPC robust against bounded unmodelled disturbances is the tube-based MPC [18]. An ancillary controller is designed in order to keep the system in an invariant tube around the trajectory generated by the nominal MPC for bounded uncertainty values. Constraints satisfaction is guaranteed if the problem is feasible. Several design strategies exist for determining suitable ancillary controllers and tube sets depending on the properties of the dynamic system. In the space domain, tube-based MPC has extensively been applied to linear dynamical systems, especially for autonomous spacecraft maneuvers [18] [31]. Even so, tube-based MPC for nonlinear systems is also an active area of research in control theory [14] [10].

This paper proposes a tube orbit predictive controller that incorporates a PINN gravity model. The structure of the paper is as follows. Section 2 introduces the orbit dynamics of a spacecraft in the vicinity of an asteroid. Section 3 develops the tube neural predictive controller. First, the PINN gravity model is integrated as part of the dynamics within the nominal MPC pipeline. Then an ancillary controller compatible with this neural dynamic system is designed to guarantee stability and constraint satisfaction for bounded gravity and solar perturbations. Section 4 illustrates the performances of the designed controller in a realistic mission scenario around 433 Eros. Section 5 concludes the paper with some remarks.

2 Spacecraft dynamics in the vicinity of asteroids

This section explains the nonlinear kinematics of the spacecraft's translational motion in orbit under the influence of an inhomogeneous gravity field and solar perturbations.

2.1 Definition of the reference frames

A rotating reference frame fixed with the asteroid $A = (O, \mathbf{i}_A, \mathbf{j}_A, \mathbf{k}_A)$ is considered. The origin O is the center of mass of the asteroid and the axes are directed along the body's principal axes. An inertial reference frame $I = (O, \mathbf{i}_I, \mathbf{j}_I, \mathbf{k}_I)$ is defined as the asteroid's body fixed frame A at the initial time. The asteroid is supposed to be in a pure uniform rotation around the $\mathbf{k}_a$ axis with an angular velocity denoted ω_a.

2.2 State-space equation

The spacecraft motion vector on the orbit in the asteroid-centered inertial (ACI) coordinates is expressed as :

$$\mathbf{r} = x\mathbf{i_i} + y\mathbf{j_i} + z\mathbf{k_i}. \tag{1}$$

By selecting the state-space vector as :

$$\mathbf{X} = \begin{pmatrix} x & y & z & \dot{x} & \dot{y} & \dot{z} \end{pmatrix}^T, \tag{2}$$

the nonlinear state space equation of the satellite motion in orbit in the ACI coordinates can be written as :

$$\dot{\mathbf{X}} = \underbrace{\begin{pmatrix} \dot{x} \\ \dot{y} \\ \dot{z} \\ -\mu\frac{x}{|\mathbf{r}|^3} \\ -\mu\frac{y}{|\mathbf{r}|^3} \\ -\mu\frac{z}{|\mathbf{r}|^3} \end{pmatrix}}_{\mathbf{F}_s(\mathbf{X})} + \underbrace{\begin{pmatrix} 0 \\ 0 \\ 0 \\ u_x \\ u_y \\ u_z \end{pmatrix}}_{\mathbf{U}_c} + \underbrace{\begin{pmatrix} 0 \\ 0 \\ 0 \\ a_{p_x} \\ a_{p_y} \\ a_{p_z} \end{pmatrix}}_{\mathbf{A}_p(\mathbf{X})}, \tag{3}$$

where μ is the asteroid's standard gravitational parameter, $a_{p_x}, a_{p_y}, a_{p_z}$ denote the perturbing accelerations on the satellite along each coordinate axis, and u_x, u_y, u_z depict the control accelerations applied by the thrusters to compensate the orbital perturbations along each coordinate axis.

In the following, we note $\mathbf{f}_s(\mathbf{r}) = \begin{pmatrix} -\mu\frac{x}{\|\mathbf{r}\|^3} \\ -\mu\frac{y}{\|\mathbf{r}\|^3} \\ -\mu\frac{z}{\|\mathbf{r}\|^3} \end{pmatrix}$, $\mathbf{u}_c = \begin{pmatrix} u_x \\ u_y \\ u_z \end{pmatrix}$, $\mathbf{a}_p(\mathbf{r}) = \begin{pmatrix} a_{p_x} \\ a_{p_y} \\ a_{p_z} \end{pmatrix}$.

2.3 Inhomogeneous gravity field

The main natural acceleration acting on a spacecraft placed in a low-asteroid orbit is the asteroid-inhomogeneous gravity field. There exists a diverse set of gravity field models.

2.3.1 Classical gravity field models

A classical approach to model a gravity field is the spherical harmonics expansion, one of the analytic solutions to Laplace's equation [3]. This model is analytically differentiable and computationally efficient if the expansion is truncated at a low degree. This is the case for quasi-spherical large bodies with first-order gravitational perturbations, like planetary oblateness. However, spherical harmonics are poorly suited to model the complex gravity field of irregular small bodies. They can struggle to capture discontinuous features like craters or mountains. [12]. Hundreds of thousands of harmonics superimposed together are usually required to represent them. Moreover, the high-order harmonic coefficients are often difficult to regress online [9]. One main reason is that homogeneously distributed samples are required to avoid the aliasing phenomena.

An alternative to the spherical harmonics model is the polyhedral gravity model. In this approach, the target body is approximated as a polyhedron comprised of facets and edges

with a homogeneous density. The exact gravity field of this polyhedron model is then expressed through the following closed-form presented by Werner and Scheeres [36]:

$$\nabla U = -G\sigma \sum_{e \in \text{ edges}} \mathbf{E}_e \cdot \mathbf{r}_e \cdot L_e + G\sigma \sum_{f \in \text{ facets}} \mathbf{F}_f \cdot \mathbf{r}_f \cdot \omega_f, \tag{4}$$

where G is the gravitational constant, σ is the density of the body, E_e is an edge dyad, r_e is the position vector between the center of the edge and the field point, L_e is an analog to the potential contribution by the edge, F_f is the face normal dyad, r_f is the distance between the face normal and the field point, and ω_f is analog to the potential contribution by the face.

The polyhedral gravity model is an interesting representation able to merge the available data on the target body in order to determine a first approximation of its gravity field. Before the start of the close-proximity orbital phase, a long in situ campaign is usually carried out to determine the body shape and its average density [16]. From these data, a polyhedral gravity model of the target body can be built. This gravity model is able to account for the irregularities of the shape model with almost exact accuracy by increasing the number of facets and edges. The remaining uncertainties in the resulting gravity field come from the constant density assumption. Even if a unique density profile can be forced onto a polyhedral gravity model, a such profile is usually unknown at this stage of the mission [32]. This well-localized source of uncertainty on the density parameter needs to be handled by a control system.

The main disadvantage of the polyhedral gravity model is its computational complexity. High-fidelity shape models of asteroids can contain hundreds of thousands of facets, which must be looped over at each propagation time step. This complexity makes it usually not compatible with the onboard system.

2.3.2 Physics-Informed Neural Network gravity model

The challenges posed by traditional gravity models motivated a new generation of gravity models that rely on machine learning methods. These models use Gaussian processes [7], extreme learning machine [6], or more recently neural networks to model the functional relationship between spacecraft position and the corresponding gravitational acceleration. These approaches demonstrated an ability to learn accurately expensive polyhedral representation while offering orders-of-magnitude improvements in terms of computational efficiency.

Recently, Martin and Schaub [11] showed that a class of neural network called PINN offers significant improvements in sample efficiency and robustness to noise by using physics priors to constrain the learned model. PINNs are able to generate accurate gravity representations using fewer and noisier measurements than the spherical harmonic model. These properties are of great interest to permit an online update of the gravity model through onboard measurements.

A PINN is a neural network that learns to approximate a system by using traditional deep learning stochastic gradient descent with a cost function that penalizes not only modeling inaccuracies but also violations of the underlying physical constraints [22].

A traditional neural network for gravity modeling would learn the mapping between the position vector and the corresponding gravity acceleration vector by minimizing a Mean-Squared Error criterion over a dataset of points $\mathcal{D} = \{(a_{G_i}, r_i)_i\}$:

$$J(\theta) = \frac{1}{N_f} \sum_{i=1}^{N_f} \left| \mathbf{a}_{Gi} - f_a \left(\mathbf{r}_i \mid \theta \right) \right|^2, \tag{5}$$

where N_f is the total number of samples, $\mathbf{a}_{Gi}$ the i-th acceleration at position $\mathbf{r}_i$ and $f_a \left(\mathbf{r}_i \mid \Theta \right)$ the output of the network given r_i and the trainable network's parameters θ which includes the weights and the biases of the network.

However, the mapping obtained through this traditional neural network may not satisfy the underlying differential equations verified by the gravity acceleration $\mathbf{a}_G$, such as:

$$\mathbf{a}_G = -\nabla U \qquad \text{Relation b/w acceleration and scalar potential U,} \tag{6a}$$

$$\nabla^2 U = 0 \qquad \text{Laplace's Equation,} \tag{6b}$$

$$\nabla \times \nabla U = 0 \qquad \text{Conservative vector field property.} \tag{6c}$$

A way to force the approximated system to remain consistent with the physics is to "Physically inform" the network by including these dynamical constraints in the cost function. To accomplish this, the network must instead learn a representation of the scalar potential $f_U \left(\mathbf{r}_i \mid \Theta \right)$, and then use automatic differentiation (AD) to take the corresponding gradient of the potential to enforce that (6a), (6b), (6c) are satisfied through the following cost function :

$$J(\Theta) = \frac{1}{N_f} \sum_{i=1}^{N_f} \left| \mathbf{a}_{Gi} - \overset{\text{AD}}{\nabla} f_U \left(\mathbf{r}_i \mid \Theta \right) \right|^2 + \left| \overset{\text{AD}^2}{\nabla} f_U \left(\mathbf{r}_i \mid \Theta \right) \right|^2 + \left| \overset{\text{AD}}{\nabla} \times \overset{\text{AD}}{\nabla} f_U \left(\mathbf{r}_i \mid \Theta \right) \right|^2. \tag{7}$$

The model-based control system developed in this paper uses, as a gravity field model, a PINN trained with this defined cost function on data collected from a polyhedral gravity model. Additionally, an online re-training of the PINN gravity model is performed in order to assess its ability to capture the density disturbances not captured by the polyhedral gravity model.

2.4 Solar perturbations

Additional natural forces acting on the spacecraft include the solar perturbations. Solar perturbations comprise the Sun third-body effect and its radiation pressure.

The Sun's third body perturbation is expressed in the asteroid-centered inertial frame as:

$$\mathbf{a}_{\text{p,sun}} \left(\mathbf{r} \right) = \mu_\odot \left(\frac{\mathbf{r}_\odot - \mathbf{r}}{\left| \mathbf{r}_\odot - \mathbf{r} \right|^3} - \frac{\mathbf{r}_\odot}{\left| \mathbf{r}_\odot \right|^3} \right), \tag{8}$$

where $\mathbf{r}_\odot$ is the Sun's position in the asteroid-centered inertial frame and $\mu_\odot$ the Sun's standard gravity parameter.

The solar radiation pressure is considered in a simplified form :

$$\mathbf{a}_{\mathrm{p,SRP}}\left(\mathbf{r}\right)=\frac{C_R P_{1AU} A}{m}\left(\frac{r_{1AU}}{\mathbf{r}_{\odot}}\right)^2\left(\frac{\mathbf{r}_{\odot}-\mathbf{r}}{|\mathbf{r}_{\odot}-\mathbf{r}|}\right), \tag{9}$$

where C_R, A, m are the spacecraft reflectivity coefficient, exposed surface and mass, r_{1AU} is the distance of one astronomical unit and P_{1AU} the solar radiation pressure at the distance of one astronomical unit [17].

3 Tube neural orbit predictive control

3.1 Onboard Neural MPC

In general, a MPC controller solves an Optimal Control Problem (OCP) by finding an input command that minimizes a cost function subject to the system dynamics while accounting for constraints.

We seek to keep the satellite on a pre-defined reference orbit with minimum tracking error and fuel consumption while guaranteeing that the control input does not exceed the maximum thrust available onboard.

Traditionally, the orbital dynamics of the spacecraft are analytically expressed by a model that often neglects the high-order gravity field perturbations as they are hard and computationally expensive to derive. In this paper, the orbital dynamics are partitioned into two terms: (1) an analytical term modeling the solar perturbations and (2) a PINN modeling the inhomogeneous gravity field. They are expressed as follows:

$$\begin{aligned}
\dot{\mathbf{X}}(t) \quad &= \mathbf{f}_{an}(\mathbf{X}(t),\mathbf{U}_c(t)) + \mathbf{f}_{PINN}(\mathbf{X}(t)),\\
\text{s.t.} \quad &\mathbf{f}_{an}(\mathbf{X}(t),\mathbf{U}_c(t)) = \mathbf{A}_{p,sun}(\mathbf{X}(t)) + \mathbf{A}_{p,SRP}(\mathbf{X}(t)) + \mathbf{U}_c(t),\\
&\mathbf{f}_{PINN}(\mathbf{X}(t)) = \mathbf{F}_s(\mathbf{X}(t)) + \mathbf{A}_{p,IG}(\mathbf{X}(t)),
\end{aligned} \tag{10}$$

where $\mathbf{X}(t)$ is the state vector, $\mathbf{A}_{p,sun}$ the sun's third body perturbation, $\mathbf{A}_{p,SRP}$ the solar radiation pressure perturbation, $\mathbf{F}_s$ the Keplerian gravity perturbation, $\mathbf{A}_{p,IG}$ the non-Keplerian gravity perturbation and $\mathbf{U}_c$ the control input.

The optimization problem that we seek to solve is :

$$\begin{aligned}
\min_{\mathbf{X}(t),\mathbf{U_c}(t)} \quad & J = \tfrac{1}{T}\int_0^T \left(\gamma\left|\mathbf{X}(t)-\mathbf{X}_r(t)\right|^2 + \left|\mathbf{U}_c(t)\right|^2\right)\mathrm{d}t,\\
\text{s.t.} \quad & \mathbf{X}(0) = \mathbf{X}_0,\\
& \dot{\mathbf{X}}(t) = \mathbf{f}_{an}(\mathbf{X}(t),\mathbf{U}_c(t)) + \mathbf{f}_{PINN}(\mathbf{X}(t)),\\
& -\mathbf{U}_{\max} \le \mathbf{U_c}(t) \le \mathbf{U}_{\max},
\end{aligned} \tag{11}$$

where $\mathbf{X}_0$ is the initial state, $\mathbf{X}_r(t)$ is the reference orbit to track, T the time horizon, $\mathbf{U}_{\max}$ the maximum control acceleration depending on the characteristics of the thrusters and γ a positive weight parameter balancing the tracking accuracy with respect to control effort.

To solve the aforementioned OCP, the system is discretized into N steps of step size Δt over the time horizon T. The state is evaluated at discrete instants $t_k = k\Delta t$ with $k \in [0, N-1]$ and the control is assumed constant within each sampling interval k. It leads to the following Nonlinear Programming Problem (NLP) problem :

$$\begin{aligned} \min_{\mathbf{X}_k, \mathbf{U}_{ck}} \quad & J = \tfrac{1}{N} \textstyle\sum_{k=0}^{N-1} \left(\gamma \left| \mathbf{X}_k - \mathbf{X}_{rk} \right|^2 + \left| \mathbf{U}_{ck} \right|^2 \right), \\ \text{s.t.} \quad & \mathbf{X}_{k=0} = \mathbf{X}_0, \\ & \mathbf{X}_{k+1} = \phi(\mathbf{X}_k, \mathbf{U}_{ck}, f, \delta t), \\ & \mathbf{f}(\mathbf{X}_k, \mathbf{U}_{ck}) = \mathbf{f}_{an}(\mathbf{X}_k, \mathbf{U}_{ck}) + \mathbf{f}_{PINN}(\mathbf{X}_k), \\ & -\mathbf{U}_{\max} \leq \mathbf{U}_{ck} \leq \mathbf{U}_{\max}, \end{aligned} \tag{12}$$

where ϕ is the numerical integration routine to discretize the dynamics equation. A fourth-order Runge-Kutta algorithm is used ($E = 4$).

The NLP problem is optimized using Sequential Quadradic Programming (SQP). This involves repetitively approximating and solving Eq.(12) as a Quadradic Programming (QP) problem. The SQP iterate is defined as $w^i = [\mathbf{X}_0^i, ..., \mathbf{X}_{N-1}^i, \mathbf{U}_{c0}^i, ..., \mathbf{U}_{cN-1}^i]$ and the step Δw^i to update the iterate, $w^{i+1} = w^i + \Delta w^i$, is given by the solution of the following QP problem :

$$\begin{aligned} \min_{\Delta \omega^i} \quad & J = \tfrac{1}{N} \textstyle\sum_{k=0}^{N-1} \begin{pmatrix} q_k^i \\ r_k^i \end{pmatrix}^\top \begin{pmatrix} \Delta \mathbf{X}_k^i \\ \Delta \mathbf{U}_{ck}^i \end{pmatrix} + \begin{pmatrix} \Delta \mathbf{X}_k^i \\ \Delta \mathbf{U}_{ck}^i \end{pmatrix}^\top H_{k,J}^i \begin{pmatrix} \Delta \mathbf{X}_k^i \\ \Delta \mathbf{U}_{ck}^i \end{pmatrix}, \\ \text{s.t.} \quad & \Delta \mathbf{X}_{k+1}^i = A_k^i \Delta \mathbf{X}_k^i + B_k^i \Delta \mathbf{U}_{ck}^i + \phi_k^i - \mathbf{X}_{k+1}^i, \forall k \in [0, N-1], \\ & -\mathbf{U}_{\max} \leq \mathbf{U}_{ck}^i + \Delta \mathbf{U}_{ck}^i \leq \mathbf{U}_{\max}, \forall k \in [0, N], \end{aligned} \tag{13}$$

where $q_k^i = \frac{\partial}{\partial \mathbf{X}_k^i} J(\mathbf{X}_k^i, \mathbf{U}_k^i)$, $r_k^i = \frac{\partial}{\partial \mathbf{U}_k^i} J(\mathbf{X}_k^i, \mathbf{U}_k^i)$, $H_{k,J}^i$ is the hessian of J in $(\mathbf{X}_k^i, \mathbf{U}_k^i)$, $\phi_k^i = \phi(\mathbf{X}_k^i, \mathbf{U}_{ck}^i, f, \partial t)$, $A_k^i = \frac{\partial}{\partial \mathbf{X}_k^i} \phi_k^i$, $B_k^i = \frac{\partial}{\partial \mathbf{U}_k^i} \phi_k^i$

The main computational burden in the solving of Eq.(13) lies in the computation of A_k^i, B_k^i, ϕ_k^i for each shooting node $k \in [0, N-1]$. It requires $N * E$ evaluations of the dynamics model $\mathbf{f} = \mathbf{f}_{an} + \mathbf{f}_{PINN}$ and $N * E * 2$ evaluations of its partial differentiations. Even if PINN gravity models are more computationally efficient than polyhedral gravity models, this method can lead, depending on the number of layers and neurons of the PINN, to extensive processing times that are not compatible with an onboard framework on a satellite.

A way to speed up the QP generation is to replace the PINN model with a locally valid approximation up to the second order around the current iterate $(\mathbf{X}_k^i, \mathbf{U}_k^i)$ of each shooting node k [24]:

$$\begin{aligned} \mathbf{f}_{PINN}(\mathbf{X}, \mathbf{U}_c) \approx \mathbf{f}_{PINN}(\mathbf{X}_k^i, \mathbf{U}_{ck}^i) + J_{k,PINN}^i \begin{pmatrix} \mathbf{X} - \mathbf{X}_k^i \\ \mathbf{U}_c - \mathbf{U}_{ck}^i \end{pmatrix} \\ + \frac{1}{2} \begin{pmatrix} \mathbf{X} - \mathbf{X}_k^i \\ \mathbf{U}_c - \mathbf{U}_{ck}^i \end{pmatrix}^\top H_{k,PINN}^i \begin{pmatrix} \mathbf{X} - \mathbf{X}_k^i \\ \mathbf{U}_c - \mathbf{U}_{ck}^i \end{pmatrix}, \end{aligned} \tag{14}$$

where $J^i_{k,PINN}$ and $H^i_{k,PINN}$ are respectively the Jacobian and the Hessian of the PINN in $(\mathbf{X}^i_k, \mathbf{U}^i_k)$. These matrices can be calculated for all shooting nodes as a single parallelized batch on central processing units (CPU) and graphics processing units (GPU) with PyTorch [21] and TensorFlow [1].

By applying Eq.(14), the QP problem becomes independent of the complexity of the PINN. However, even with a computationally light dynamics model, solving a sequence of QP until convergence is usually not compatible with an onboard framework on a satellite. Therefore, we adopt in this paper a real-time iteration scheme (RTI) [5], where a single iteration of the SQP is executed.

3.2 Tube neural MPC: Boundary layer sliding mode ancillary controller

Unmodelled solar or gravitational disturbances can affect the optimality and accuracy of the neural MPC, and lead the spacecraft to escape the nominal trajectory. In this paper, a tube-based MPC approach is proposed in order to guarantee the robustness of the designed controller against bounded disturbances.

3.2.1 Principle of a tube MPC

To take into account these unmodelled disturbances in the control system, the following continuous-time uncertain dynamics system is considered:

$$\ddot{\mathbf{r}} = \underbrace{\begin{pmatrix} -\mu\frac{x}{|\mathbf{r}|^3} \\ -\mu\frac{y}{|\mathbf{r}|^3} \\ -\mu\frac{y}{|\mathbf{r}|^3} \end{pmatrix}}_{\mathbf{f}_s(\mathbf{r})} + \underbrace{\begin{pmatrix} u_x \\ u_y \\ u_z \end{pmatrix}}_{\mathbf{u}_c} + \underbrace{\begin{pmatrix} a_{p_x} \\ a_{p_y} \\ a_{p_z} \end{pmatrix}}_{\mathbf{a}_p(\mathbf{r})} + \underbrace{\begin{pmatrix} d_x \\ d_y \\ d_z \end{pmatrix}}_{\mathbf{d}}, \tag{15}$$

where d_x, d_y, d_z are unknown external disturbances. We assume that d_x, d_y, d_z are bounded: $\exists B \in \mathbb{R}$ such that $|d_x|, |d_y|, |d_z| < B$.

As in a standard tube MPC approach, a two-level control design is established for the control input of the uncertain dynamics system Eq.(15):

$$\mathbf{u}_c = \mathbf{u}_n + \mathbf{u}_a, \tag{16}$$

where $\mathbf{u}_n$ is the control input of the nominal MPC based on the ideal system without unmodelled disturbances and $\mathbf{u}_a$ the control input of an ancillary controller that seeks to return the perturbed state to the nominal state by counteracting the unmodelled disturbances. For bounded disturbances, this ancillary controller brings the state in a tube set centered on the nominal state.

3.2.2 Design of the ancillary controller

An ancillary controller that has been extensively used for non-linear tube MPC for its simplicity and robustness is a sliding mode controller. Unlike other control strategies, sliding

mode control completely cancels any bounded modeling error. However, complete cancellation comes at the cost of high-frequency discontinuous control making it impractical for satellite operations. A number of versions that ensure continuity in the control signal have been developed. In this paper, a boundary layer sliding mode controller (BLSMC), originally developed by Slotine and Sastry [30], is used as an ancillary controller.

The aim of the ancillary controller is to force the real system to track the nominal system. In this regard, the following tracking error is defined:

$$\mathbf{e} = \mathbf{r} - \mathbf{r}_*, \tag{17}$$

where $\mathbf{r}$ is the real position of the system and $\mathbf{r}_*$ the nominal position of the system. As in a classic sliding mode control approach, the controller is designed with a sliding variable $\mathbf{s}$ function of the tracking error $\mathbf{e}$:

$$\mathbf{s} = \begin{pmatrix} s_x \\ s_y \\ s_z \end{pmatrix} = \left(\frac{d}{dt} + \lambda\right)\mathbf{e} = \dot{\mathbf{e}} + \lambda\mathbf{e}, \tag{18}$$

where λ is a positive constant. A sliding manifold $\mathcal{S}$ is defined such that $\mathbf{s} = 0$. Remaining on this sliding surface $\mathcal{S}$ guarantees that the tracking error $\mathbf{e}$ goes to zero exponentially.

A sufficient condition to make this sliding surface $\mathcal{S}$ reachable in finite time and invariant is to satisfy outside $\mathcal{S}$:

$$\forall i \in \{x, y, z\}, \quad \frac{1}{2}\frac{d}{dt}s_i^2 \leq -\eta\left|s_i\right|, \tag{19}$$

where η is a positive constant dictating the convergence rate to the sliding surface.

Differentiating Eq.(18) with respect to time results in:

$$\begin{aligned} \dot{\mathbf{s}} &= \ddot{\mathbf{e}} + \lambda\dot{\mathbf{e}}, \\ &= \ddot{\mathbf{r}} - \ddot{\mathbf{r}}_* + \lambda\dot{\mathbf{e}}, \\ &= (\mathbf{f}_s(\mathbf{r}) + \mathbf{a}_p(\mathbf{r}) + \mathbf{u}_c + \mathbf{d}) - (\mathbf{f}_s(\mathbf{r}_*) + \mathbf{a}_p(\mathbf{r}_*) + \mathbf{u}_{MPC}) + \lambda\dot{\mathbf{e}}, \\ &= (\mathbf{f}_s(\mathbf{r}) - \mathbf{f}_s(\mathbf{r}_*)) + (\mathbf{a}_p(\mathbf{r}) - \mathbf{a}_p(\mathbf{r}_*)) + \mathbf{d} + \mathbf{u}_a + \lambda\dot{\mathbf{e}}. \end{aligned} \tag{20}$$

A control law $\mathbf{u}_a$ that satisfies the sliding condition Eq.(19) is:

$$\mathbf{u}_a = -\left(\mathbf{f}_s(\mathbf{r}) - \mathbf{f}_s(\mathbf{r}_*)\right) - \left(\mathbf{a}_p(\mathbf{r}) - \mathbf{a}_p(\mathbf{r}_*)\right) - \lambda\dot{\mathbf{e}} - K\mathrm{sgn}(\mathbf{s}), \tag{21}$$

where $K = B + \eta$, B being the upper bound of the element-wise unknown external disturbances.

As shown in Eq.(21), the control laws that satisfy the sliding condition Eq.(19) are discontinuous across the surface $\mathcal{S}$, thus leading to control chattering. It is a phenomenon highly undesirable in orbit control since it involves extremely high control activity that may excite unmodeled high-frequency dynamics and shorten actuators' life spans.

Slotine and Sastry [30] suggested a solution to this problem by smoothing out the control discontinuity in a thin boundary layer neighboring the switching surface $\mathcal{S}$:

$$\mathcal{B} = \{\mathbf{r} : |s_x| \leq \Phi, |s_y| \leq \Phi, |s_z| \leq \Phi\}, \tag{22}$$

where Φ is a time-independent boundary layer thickness. This is achieved by choosing a control law $\mathbf{u}_a$ similar as Eq.(21) outside $\mathcal{B}$ and then interpolating $\mathbf{u}_a$ inside $\mathcal{B}$ by replacing the discontinuous term $\text{sgn}(\mathbf{s})$ by $\mathbf{s}/\Phi$.

The control law $\mathbf{u}_a$ becomes:

$$\mathbf{u}_a = -\left(\mathbf{f}_s(\mathbf{r}) - \mathbf{f}_s(\mathbf{r}_*)\right) - \left(\mathbf{a}_c(\mathbf{r}) - \mathbf{a}_c(\mathbf{r}_*)\right) - \lambda\dot{\mathbf{e}} - K\text{sat}(\mathbf{s}/\Phi), \tag{23}$$

where $K = B + \eta$ and sat is the element-wise saturation function.

This leads to a control law that tracks the nominal trajectory within a guaranteed precision Ω while allowing avoidance of control chattering. This guaranteed precision Ω represents the radius of the tube around the nominal trajectory and can be determined by integrating Eq.(18):

$$\forall i \in \{x, y, z\}, \quad e_i(t) = e^{-\lambda t} e_i(0) + \int_0^t e^{-\lambda(t-\tau)} s_i d\tau \tag{24}$$

By assuming that $e(0) = 0$:

$$\forall i \in \{x, y, z\}, \quad |e_i(t)| = \left| \int_0^t e^{-\lambda(t-\tau)} s_i d\tau \right| \leq \frac{\Phi}{\lambda} = \Omega \tag{25}$$

3.2.3 Constraint tightening

The actuator constraint in the optimization problem solved by the nominal MPC:

$$-\mathbf{u}_{\max} \leq \mathbf{u_c} \leq \mathbf{u}_{\max}, \tag{26}$$

must be modified to account for the control input from the ancillary controller caused by model errors and disturbances.

The tracking error bound Eq.(25) and the sliding variable bound Eq.(22) can be used to obtain an upper bound on the absolute value of the control input from the ancillary controller:

$$\begin{aligned} |\mathbf{u}_a| &= |-\left(\mathbf{f}_s(\mathbf{r}) - \mathbf{f}_s(\mathbf{r}_*)\right) - \left(\mathbf{a}_p(\mathbf{r}) - \mathbf{a}_p(\mathbf{r}_*)\right) - \lambda\dot{\mathbf{e}} - Ksat(\mathbf{s}/\Phi)|, \\ &\leq \max_{\mathbf{r}_* - \Omega\mathbf{i} \leq \mathbf{r} \leq \mathbf{r}_* + \Omega\mathbf{i}} \left(|\mathbf{f}_s(\mathbf{r}_*) + \mathbf{a}_p(\mathbf{r}_*) - \left(\mathbf{f}_s(\mathbf{r}) + \mathbf{a}_p(\mathbf{r})\right)|\right) + \lambda\left(\Phi\mathbf{i} + \lambda\Omega\mathbf{i}\right) + K\mathbf{i}, \\ &\leq D + \lambda\left(\Phi\mathbf{i} + \lambda\Omega\mathbf{i}\right) + K\mathbf{i}, \\ &= \mathbf{b}_a, \end{aligned} \tag{27}$$

where max is the element-wise maximum function, $|.|$ is the element-wise absolute value function, D a constant upper bound of the maximized term and $\mathbf{i} = \begin{pmatrix} 1 & 1 & 1 \end{pmatrix}^T$.

The bound established in Eq.(27) can be used to tighten the actuator constraint:

$$\forall i \in \{x, y, z\}, \quad |u_{c,i} - u_{a,i}| = |u_{n,i}| \leq u_{\max,i} - b_{a,i}. \tag{28}$$

4 Case Study: Unstable polar orbit around 433 Eros

In this section, the performances of the tube orbit predictive controller are illustrated in a realistic mission scenario around 433 Eros. First, the scenario key variables such as the target asteroid, the target orbit, and the spacecraft characteristics are declared. Then, the PINN gravity model, later integrated into the dynamic model of the controller, is trained to learn the known gravity field of 433 Eros. Finally, the proposed control strategy is used to track an unstable polar orbit with dynamic uncertainties and the numerical results are shown.

4.1 Mission scenario

4.1.1 Target asteroid

The target asteroid is 433 Eros, a benchmark for small-body missions. Its irregular and elongated shape contributes to a highly inhomogeneous gravitational field and therefore offers a challenging environment for orbital operations. Thanks to the NEAR Shoemaker mission, high-resolution shape models and numerous physical characteristics are available. 433 Eros is a S-type asteroid with a size of $33 \times 13 \times 13$ km^3, a gravitational parameter of $\mu = 4.4628 \cdot 10^5$ m^3 s^{-2}, an average density of $\sigma = 2.67$ g cm^{-3} and a rotation period of $T = 5.27$ h around its major inertia axis. 433 Eros orbits around the Sun with a semi-major axis $a_s = 1.46$ AU. For simplicity, we assume the Sun position to be constant, equal to $\mathbf{r}_\odot = \begin{pmatrix} a_s & 0 & 0 \end{pmatrix}^T$ AU.

For the real-dynamics simulation, the gravitational field of 433 Eros is represented by a polyhedral model with 5144 facets and a realistic and heterogeneous density profile. This density profile is built by dividing the polyhedron model into 16061 tetrahedral and distinguishable elements and assigning a density value to each of them according to a multivariate normal distribution. This distribution is characterized by a mean vector that precises the average density value of each element and a covariance matrix that specifies the covariance value between all the possible combinations of elements. The average density value of each element is set to σ and the covariance matrix is calculated with the help of the Matérn Kernel [2]. The Matérn Kernel is a parametric covariance function that specifies the covariance between two elements only as a function of the distance between their centers of mass r:

$$\mathbf{M}(r) = \frac{\sigma^2}{\Gamma(\nu) 2^{v-1}} \left(\frac{2\sqrt{\nu}}{l} r \right)^{\nu} \mathcal{K}(\nu) \left(\frac{2\sqrt{\nu}}{l} r \right), \tag{29}$$

where Γ is the Gamma function, $\mathcal{K}$ the Bessel function, σ^2 the variance parameter, ν the smoothness parameter, and l the scaling parameter. The density distribution adopted for the real-dynamics simulation is defined using a variance parameter of 300^2, a smoothness parameter of 1.5, and a scaling parameter of 2. The obtained density profile can be visualized in Figure 1.

In this mission scenario, we assume that the solar perturbations around 433 Eros and its shape model are known before the start of the mission, but not its density profile. The known gravity field is represented by a polyhedral model also with 5144 facets, but a homogeneous density equal to σ. The in-homogeneous density profile represents an unmodelled disturbance.

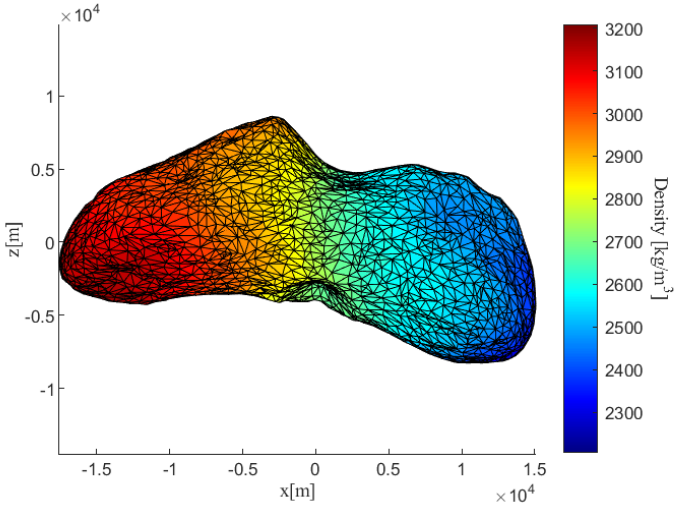


Figure 1: Density profile of the polyhedron model for the real-dynamics simulation ($\sigma^2 = 300^2$, $\nu = 1.5$, $l = 2$).

4.1.2 Target orbit

Polar orbits and the global coverage they provide are of particular interest to any asteroid mission. During the NEAR Shoemaker mission, the spacecraft spent an extended period in quasi-circular polar orbits with a radius ranging from 35 to 50 km. There was an interest in decreasing the orbit radius as much as possible to gain additional signals in the gravitational measurements. However, during the mission, it was numerically determined that polar orbits had a lower limit radius roughly equal to 35 km, that if violated, led to unstable orbital motion [27].

The proposed controller aims to stabilize the motion of the spacecraft on circular polar orbits of radius inferior to this limit, despite dynamic uncertainties. In this mission scenario, a polar orbit of radius $R = 23$ km is considered. Figure 2 shows the unstable motion of an uncontrolled spacecraft initially on this orbit. The instabilities lead the spacecraft to impact the asteroid after 26.5 h and 3 orbital revolutions.

4.1.3 Spacecraft characteristics

In this mission scenario, the design of the spacecraft is based on Hayabusa 2, a Japanese spacecraft that explored asteroid Ryugu (162173) from June 2018 to November 2019. Similarly, it has a wet mass m_0 of 600 kg, a solar radiation pressure exposed area S of 13.276 m^2 and a coefficient of reflectivity C_R equal to 1.321. The spacecraft also features three electric thrusters. It is a suitable propulsion method for the continuous orbital control considered in this mission. No bi-propellant reaction control subsystem is considered in this paper as an adjunct for trajectory control. Differently from Hayabusa 2, the electric Xenon thrusters are chosen with properties similar to the newly developed Advanced Electric Propulsion System (AEPS). They are characterised by an I_{sp} of 2900 s

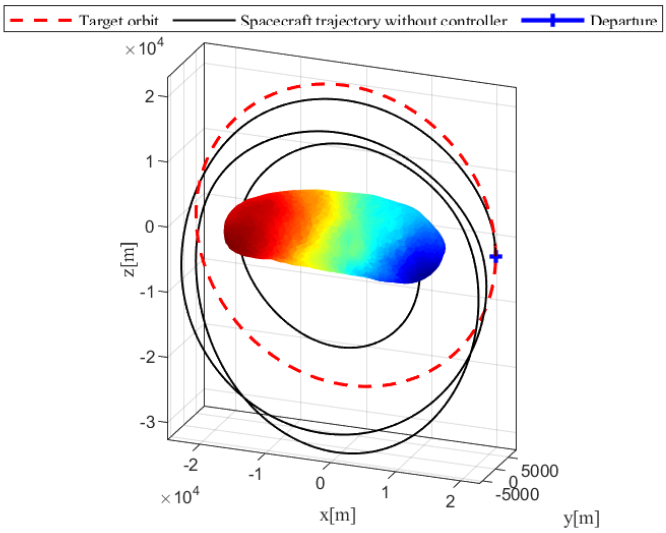


Figure 2: Propagation of a polar orbit in the inertial frame for 26.5 h with an initial radius of 23 km.

and a maximum thrust T of 600 mN. It is assumed that a 30 kg Xenon tank is dedicated to this mission. The fuel consumption m_F of the spacecraft is measured as:

$$m_F \approx \int_0^{T_{sim}} \frac{m_0 \left|\mathbf{a}_u(t)\right|}{g_0 I_{\mathrm{sp}}} \mathrm{d}t, \tag{30}$$

where T_{sim} is the simulation time, $|\mathbf{a}_u(t)|$ the norm of the control acceleration and g_0 the standard gravity on Earth.

4.2 PINN dynamic model training

The PINN presented in section 2.3.2 is employed to learn accurately the partially known gravitational field of 433 Eros, represented by a polyhedral gravity model with 5144 facets and a homogeneous density $\sigma = 2.67\ \mathrm{g\,cm^{-3}}$. This PINN is then integrated into the dynamic model of the orbit controller.

4.2.1 Training data

In order to train the PINN to learn the functional relationship between positions about 433 Eros and the corresponding inhomogeneous gravity acceleration, a data set of $N_f = 10^5$ input-response pairs is assembled.

The general process of generating the i-th pair consists of five steps:

1. A position vector $\mathbf{r}_i$ is sampled from a uniform distribution within a sphere centered at the center of mass of the asteroid and with a radius R_{sphere} equal to 60 km.
2. The gravity potential $U_{poly}(\mathbf{r}_i)$ and acceleration $\mathbf{a}_{poly}(\mathbf{r}_i)$ are obtained from the polyhedral model.

3. The following criterion is employed to discard the sample if its location is inside the asteroid's surface :

$$\nabla^2 U_{poly}(\mathbf{r}_i) = \begin{cases} -4\pi\sigma G, & \text{if } \mathbf{r}_i \text{ is inside the asteroid} \\ 0 & \text{if } \mathbf{r}_i \text{ is outside the asteroid} \end{cases} \tag{31}$$

where σ is the constant density of the polyhedron model and G the gravitational constant.

4. $(\mathbf{r}_i, \mathbf{a}_{poly}(\mathbf{r}_i))$ is saved as a pair in the training batch. $U_{poly}(\mathbf{r}_i)$ is kept for the normalization of the training data.

4.2.2 Normalization of the training data

In machine learning applications, the training data are usually normalized to ensure that the weights and biases learned by the network are not exceeding large magnitude, with the goal of reducing the learning complexity and avoiding large numerical errors [8]. In this paper, the following physically relevant nondimensionalization are introduced :

$$\mathbf{r}^\star = \frac{\mathbf{r} - r_0\mathbf{i}}{r_s}, \tag{32a}$$

$$U^\star_{poly} = \frac{U_{poly} - U_0}{U_s}, \tag{32b}$$

$$\mathbf{a}^\star_{poly} = -\frac{dU^\star_{poly}}{d\mathbf{r}^\star} = -\frac{r_s}{U_s}\mathbf{a}_{poly}, \tag{32c}$$

where r_0 is the minimum value of all position components in the training dataset, r_s the range of the position vector, U_0 the minimum of the potential, U_s the range of the potential and $\mathbf{i} = \begin{pmatrix} 1 & 1 & 1 \end{pmatrix}^T$.

4.2.3 Hyperparameters for the training

The PINN used for training is densely connected with 8 hidden layers and 20 neurons per layer. This structure offers a desirable balance between network capacity and reasonable training time.

For each hidden layer, the Gaussian exponential linear unit (GELU) is used as an activation function. This function was chosen instead of the popular Rectified linear unit (ReLU) for its infinite orders of continuity, especially important for the training of a PINN. If the activation function does not have smooth high-order derivatives, and if the gradients of the network are taken using automatic differentiation to enforce the physics constraints, the cost function will no longer be well-behaved for gradient descent [12].

The PINN is trained by mini-batches of 1000 samples for a number of epochs set to 8000 with the Adam optimizer. We also adopt an early stopping procedure to halt the training in case of overfitting (by checking the validation loss over a batch of 10^4 samples, with a patience of 500 epochs and a minimum delta of 0).

The learning rate is initially set to $\eta_0 = 0.005$ and then decays exponentially over time to prevent oscillations around the local minimum. The following exponentially decaying

learning rate η_i is applied at epoch i :

$$\eta_i = \begin{cases} \eta_0 & i < i_0, \\ \eta_0 \cdot \alpha^{(-\frac{i-i_0}{s})} & i \geq i_0, \end{cases} \tag{33}$$

where $i_0 = 2000$ is the reference epoch after which the decay begins, $s = 8000$ the scale parameter and $\alpha = 0.01$ the decay rate.

All the hyperparameters used for training are summarized in Table 1.

Hyperparameter	Value
Number of layers	8
Nodes per layer	20
Activation function	GELU
Weight initialization	Glorot normal
Epochs	8000
Mini-batch size	1000
Optimizer	Adam
η_0	0.005
i_0	200
α	0.01
s	8000

Table 1: Hyperparameters for the training of the PINN.

4.2.4 Results of the training

The PINN was trained with PyTorch on a laptop with 8 GB of RAM and an Intel Core i7-1065G7 processor running at 3.9 GHz. The training lasted $T_{train} = 1.83$ h and was interrupted by the early stopping function at the 6127th epoch.

The performances of the PINN are assessed on a validation dataset of size 10^4 and generated in the same way as the training dataset. The regression analysis on this validation dataset reported in Figure 3 validates the learning process.

The Pearson Correlation Coefficients greater than 0.9999 and the regression lines close to the identity function confirm the ability of the PINN to predict the gravity accelerations for position vectors not included in the training dataset.

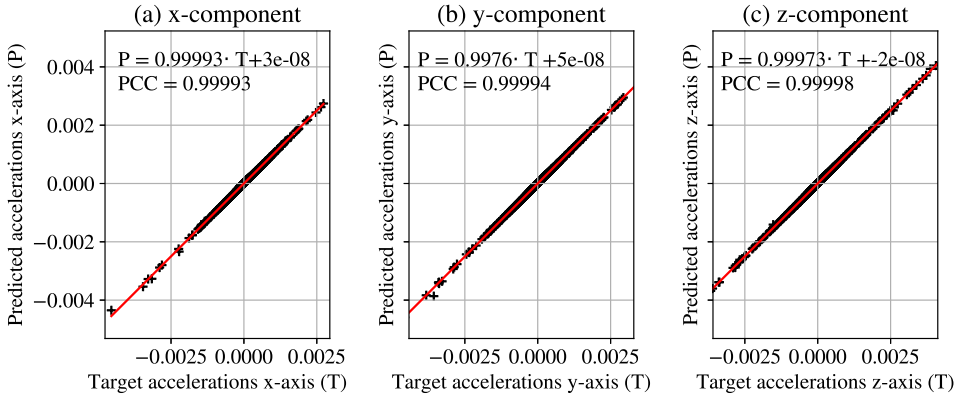


Figure 3: Regression plots on 10^4 validation field points, reporting the target gravity attraction of the constant polyhedron in $\mathrm{m\,s^{-1}}$ compared with the one predicted by the PINN in $\mathrm{m\,s^{-1}}$. (PCC = Pearson Correlation Coefficient)

To quantify the error between the PINN gravity model and the polyhedral gravity model, the acceleration residuals for the validation dataset are reported in Figure 4.

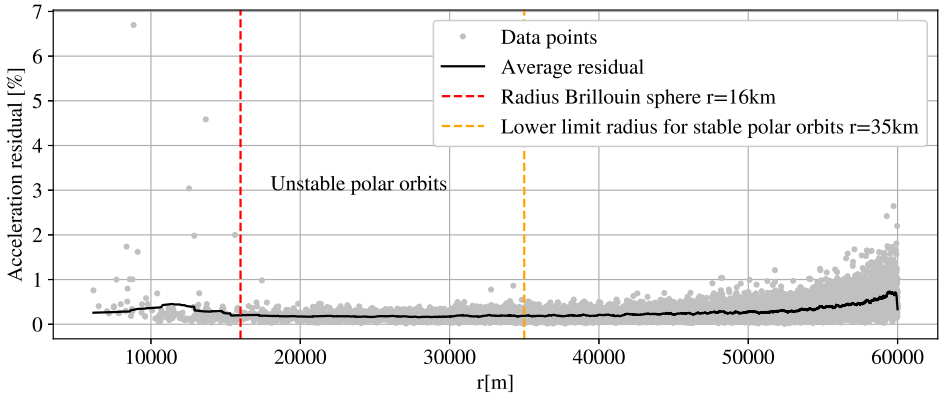


Figure 4: Acceleration residuals, $|\mathbf{a}_{poly} - \mathbf{a}_{PINN}|/|\mathbf{a}_{poly}|$, for 10^4 validation field points as a function of the radius.

On the unstable polar orbits outside the Brillouin sphere, the PINN is able to predict the gravity acceleration with less than 0.2% error on average. This performance could be improved by increasing the size of the training dataset. However, in this case study, this error is negligible with respect to the uncertainties on the gravity field of Eros before the mission. As a matter of comparison, the acceleration uncertainty $|\mathbf{a}_{poly} - \mathbf{a}_{real}|/|\mathbf{a}_{poly}|$ at a 23 km radius from the center of mass of the asteroid has an average of 4.1%, so 20 times higher than the average error of 0.2% introduced by the PINN.

The mean evaluation time of the target polyhedron gravity model and the trained PINN on 1000 field points are compared in Table 2. We notice a division by 15 of the evaluation time.

Gravity model	Evaluation time [s]
Target polyhedron model (homogeneous density, 5144 facets)	0.0774
Trained PINN	0.0052

Table 2: Comparison of the mean evaluation time of one field point for the target polyhedral gravity model and the trained PINN.

4.3 Performance of the tube-MPC controller with dynamic uncertainties

In this section, a simulation is performed to assess the ability of the designed controller to track the target unstable orbit with dynamic uncertainties. It is assumed that the navigation system is able to estimate the exact state of the spacecraft at the chosen control frequency.

4.3.1 Simulation parameters

The designed controller was implemented in Python on a laptop with 8 GB of RAM and an Intel Core i7-1065G7 processor running at 3.9 GHz.

The simulation lasts one orbital period of the target orbit noted T_{orbit} and the control frequency f is set to 1 Hz.

Before the start of the simulation, the onboard neural MPC defined in section 3.1 generates a nominal trajectory and nominal control inputs by solving T_{orbit}/f times the NLP problem 12. This MPC is not run in real-time but is designed to be re-run onboard each time the dynamic model is updated. Concerning the definition of the NLP problem, the tracking error weight γ is chosen to give a high priority to reduce tracking error with respect to fuel consumption. The maximum thrust vector $\mathbf{U}_{max}$ is set in accordance with the thrusters' characteristics defined in section 4.1.3 and the maximum thrust constraint is tightened following section 3.2.3. To speed up the solving of the NLP problems, the PINN gravity model is linearized to the first order around the current iterate of each shooting node by using the ML-Casadi framework developed in [24]. For a 1 Hz control frequency, this computational simplification induces an error negligible with respect to the uncertainties on the gravity model of Eros.

While simulating, the BLSM ancillary controller defined in section 3.2.2 is run in real-time to keep the real trajectory within a tube of radius Ω around the nominal trajectory by counteracting the unmodelled gravity disturbances. The upper disturbance bound B is validated by plotting on Figure 5 the error between the PINN gravity model and the real world gravity model defined in section 4.1.1 on a 23 km sphere around Eros.

All the simulation parameters are summarized in Table 3.

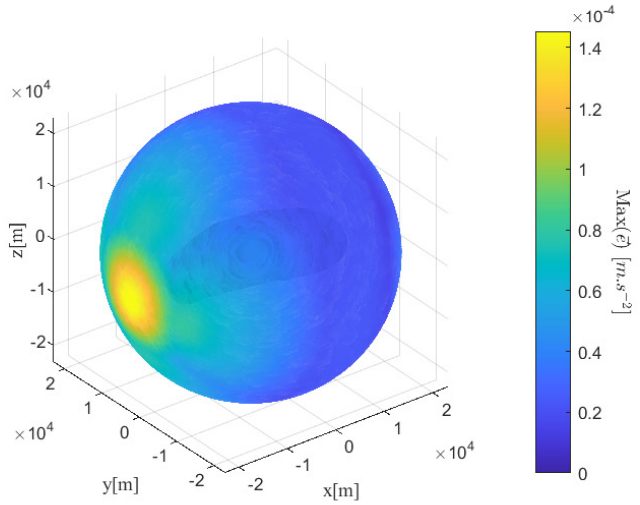


Figure 5: Maximum component of the disturbance vector $\mathbf{e} = |\mathbf{a}_{PINN} - \mathbf{a}_{Real}|$ with respect to the position on a 23km sphere around the asteroid.

Parameters	Value
Simulation time T_{orbit}	9.113 h (Period of a polar orbit at 23km)
Initial state vector	$\begin{pmatrix} 23\times10^3\,\mathrm{m} & 0\,\mathrm{m} & 0\,\mathrm{m} & 0\,\mathrm{m\,s^{-1}} & 0\,\mathrm{m\,s^{-1}} & \sqrt{\frac{\mu}{(23\cdot10^3)^3}}\,\mathrm{m\,s^{-1}} \end{pmatrix}^T$
Control frequency	1Hz
Onboard MPC	
Time horizon T	60 s
Number of steps N	60
Weight parameter γ	10^3
Maximum thrust $\mathbf{U}_{max}$	$\begin{pmatrix} 1 & 1 & 1 \end{pmatrix}^T * 10^{-3}\ \mathrm{m\,s^{-2}}$
D	$10^{-6}\mathrm{m\,s^{-2}}$
Order PINN approximation	1
Ancillary controller	
λ	$0.01\ \mathrm{s^{-1}}$
η	$1\cdot10^{-5}\ \mathrm{s^{-2}}$
Upper bound disturbance B	$1.5\cdot10^{-4}\ \mathrm{m\,s^{-2}}$
Radius of the tube Ω	0.5 m

Table 3: Parameters of the tube-MPC.

4.3.2 Results of the simulation

The nominal trajectory and control inputs are generated at a mean frequency of 52.6 Hz. The nominal trajectory is tracking the reference trajectory with an error inferior to 10^{-6} m, which is in accordance with the definition of the tracking error weight. The nominal control inputs are displayed in Figure 6. They stabilize the orbit by counteracting the modeled non-keplerian solar and gravity forces acting on the spacecraft. They satisfy the

defined tightened constraint and lead to a nominal fuel consumption equal to $8.298 \cdot 10^{-2}$ kg.

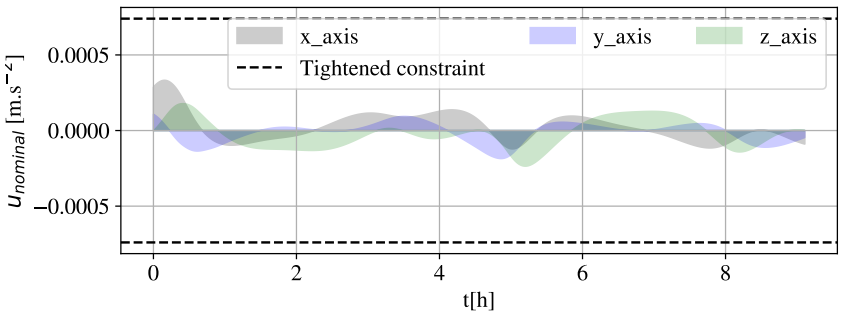


Figure 6: Nominal control inputs.

The ancillary controller is run in real-time to counteract the unmodelled non-keplerian forces acting on the spacecraft. These unmodelled forces along the nominal trajectory are displayed in Figure 7.

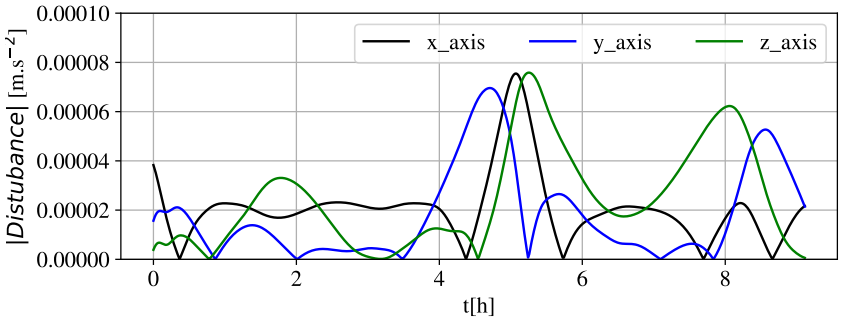


Figure 7: Absolute value of the unmodelled disturbances along the nominal orbit.

The mean time to generate a control input is 0.0124 s, which is consistent with the control frequency. The control inputs and the error between the nominal and real trajectories are shown in Figure 8. The nominal control inputs are slightly modified by the ancillary controller in order to keep the real trajectory in a 0.5 m tube around the nominal trajectory. Thanks to the constraint tightening performed on the nominal MPC, the control inputs are compatible with the maximum thrust available onboard. An increase in the fuel consumption of 4.41% with respect to the nominal case is observed.

This simulation illustrates the robustness of the designed controller against bounded disturbances. The next section investigates the possibility of updating online the dynamic model of the onboard neural MPC to enhance the optimality of the nominal control inputs.

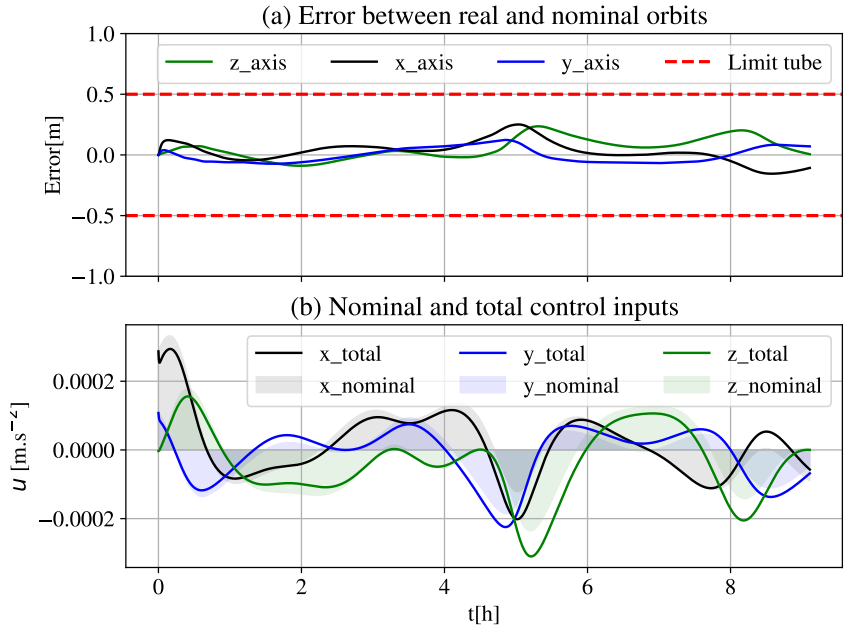


Figure 8: Error between the real-time and nominal orbits, and total control inputs.

4.4 Optimization of the performance using online training of the PINN

4.4.1 Retraining of the PINN

To retrain the PINN gravity model during a mission, online measurements of the spacecraft position and the corresponding inhomogeneous gravitational acceleration at that position must be gathered. A way to provide these training data is to estimate directly the unmodelled gravity accelerations as part of the state of the onboard navigation filter using high-order forms of dynamic model compensation. In line with the performances of this technique illustrated by Sanchez and Schaub [25], this experiment assumes that erroneous gravity accelerations can be calculated and stored once every 10 minutes during the mission. The noisy acceleration estimates are generated by first calculating the true acceleration at the current spacecraft position, and then adding an error vector with a random direction and a random magnitude up to 10% of the true acceleration magnitude. It is worth emphasizing that this choice results in higher acceleration errors when the spacecraft is closer to the asteroid than when it is far away. While this choice is not necessarily reflective of true filter estimates and evolving covariances, it remains useful as it provides challenging training data that can be used to highlight the strengths and weaknesses of online re-training.

The training data are not collected on the unstable 23 km target orbit but on four circular stable polar orbits of radius 35, 40, 45, and 50 km. This data-gathering process counts on

the ability of the PINN to predict the gravity field on the target orbit from training data collected on stable orbits. In this way, the performance of the controller can be optimized from the start of the operation on the target orbit. On each stable orbit, the spacecraft is collecting data for 15 days with a cadence of 10 minutes, which leads to 8644 samples.

The PINN gravity model, previously initialized on the homogeneous polyhedral model of Eros in section 4.2, is retrained on this dataset with the hyperparameters displayed in Table 4.

Hyperparameter	Value
Size training dataset	8644
Mini-batch	No
Epochs	400
Constant learning rate	10^{-5}
Optimizer	Adam

Table 4: Hyperparameters for the retraining of the PINN.

The retraining lasted $T_{retrain} = 14$ s. The error between the retrained PINN gravity model and the real-world gravity model on a 23 km sphere around Eros is shown in Figure 9.

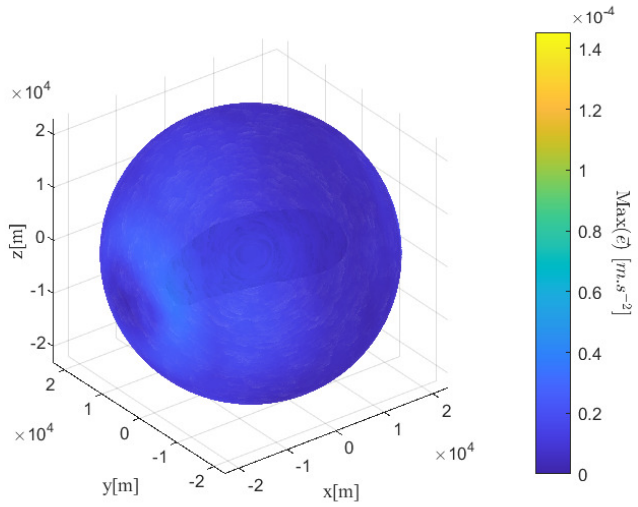


Figure 9: Maximum component of the disturbance vector $\mathbf{e} = |\mathbf{a}_{PINN} - \mathbf{a}_{Real}|$ with respect to the position on a 23 km sphere around the asteroid after retraining of the PINN.

In comparison with Figure 5, a 76.4% decrease in the maximum error on the 23 km sphere around the asteroid, is noticed after the retraining of the PINN. This result confirms the predicting capacities of the PINN for density distribution disturbances.

4.4.2 Performance of the tube-MPC after retraining of the PINN

After the online retraining of the PINN, the dynamic model of the neural MPC is updated. New nominal control inputs and a new nominal trajectory are generated. The tracking error between the nominal and target orbits is not exceeding 10^{-6} m as before the retraining and the nominal fuel consumption is estimated at $8,548 \cdot 10^{-2}$ kg.

The real-time simulation is run again with the same simulation parameters as in section 4.3.1. The unmodelled forces along the nominal trajectory are displayed in Figure 10.

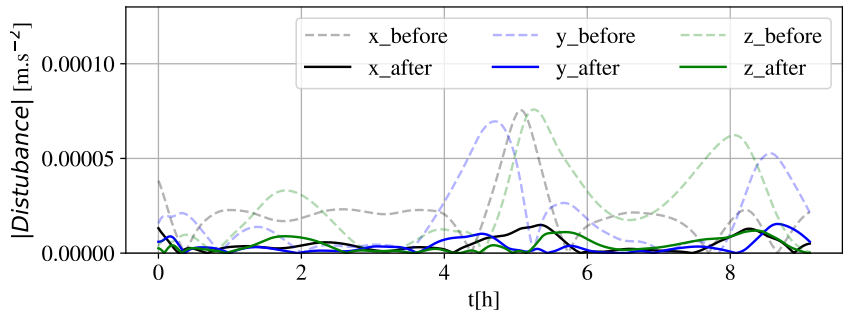


Figure 10: Absolute value of the unmodelled disturbances along the nominal orbit before and after the retraining of the PINN.

The control inputs and the error between the nominal and real trajectories are shown in Figure 11. Thanks to the online retraining, the real trajectory and control inputs converge toward the nominal case. The maximum element-wise tracking error between the nominal and real trajectories decreased by 66.6%. Additionally, the maximum element-wise control input from the ancillary controller was reduced by 66.4%. With respect to the nominal case, an increase in the fuel consumption of 1.36% is noticed, instead of 4.41% before retraining.

The online retraining reduces the magnitude of the control inputs from the ancillary controller and allows the tube-MPC to converge towards an optimal controller.

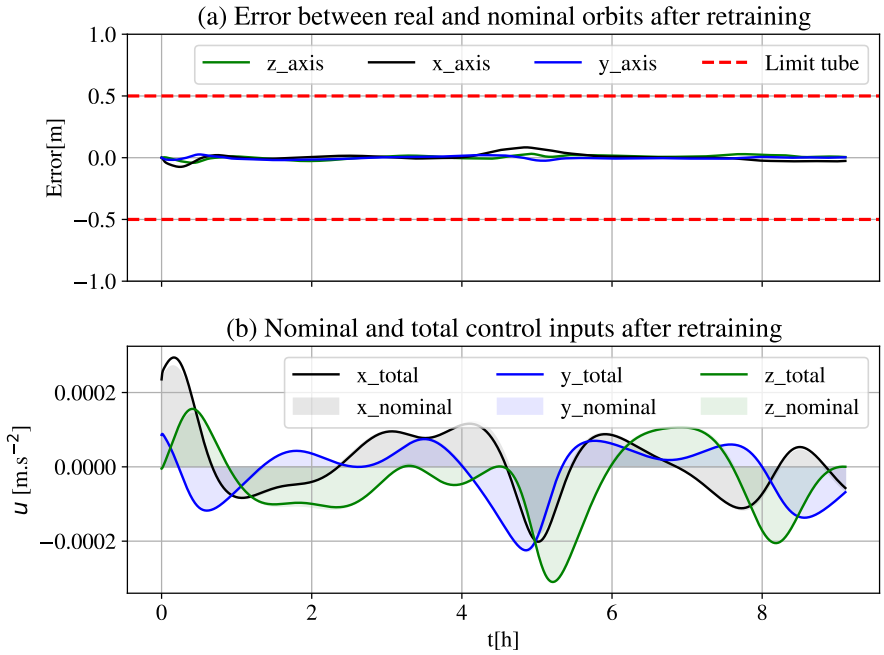


Figure 11: Error between the real-time and nominal orbits, and total control inputs after retraining of the PINN.

5 Conclusion

This paper presents a tube neural predictive controller for orbital control around asteroids. The main ingredients are: (1) a PINN that models in a computationally efficient way the complex asteroid's gravity field; (2) a neural MPC that based on this PINN generates optimal control inputs and trajectory to track the target orbit; (3) an ancillary controller that counteracts online the unmodelled disturbances. For bounded disturbances in the dynamic model, the designed controller guarantees that the spacecraft will remain within a tube around the optimal trajectory. The radius of the tube is user-chosen, depending on the aggressiveness of the control allowed by the actuators. Moreover, we experimentally validate that a retraining of the PINN gravity model on noisy data collected online can reduce the magnitude of unmodelled disturbances and optimize the performances of the designed controller.

Future work consists of including this designed controller in a full guidance, navigation, and control system. First, the tracking error guarantees given by the tube approach can be changed to take into account the navigation filter uncertainties. Then, the gravity disturbances can be estimated online as part of the state of the navigation filter using high-order forms of dynamic model compensation. Thus, a dataset for the online retraining of the PINN gravity model can be built.

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